

BUILDING ENERGY PERFORMANCE THROUGH MACHINE LEARNING

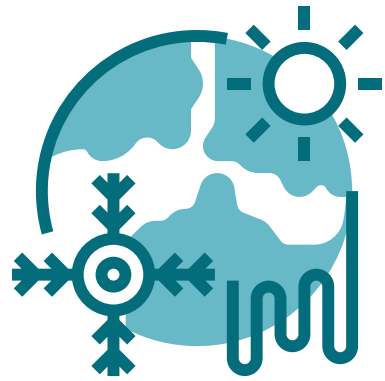
A comprehensive study investigating how constructive variables of building envelopes impact heating and cooling demands across two distinct architectural shapes in southern Brazil's Köppen Cfa climate.



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INTRODUCTION



CLIMATE CHANGE

Climatic emergency has sparked global debates on policies, strategies, and goals for mitigating their effects.

[IPCC, 2022]



ENERGY EFFICIENCY IN BUILDINGS

In 2022, buildings contributed to around 39% of greenhouse gas emissions and accounted for roughly 35% of global energy consumption.

[IEA, 2023]



PARAMETRIC SIMULATIONS

Algorithms, simulations, and parametric programs streamline collaboration between architecture, engineering, and computing, cutting down the time needed to find optimal solutions.

[ATTIA et al., 2013; ØSTERGÅRD; JENSEN; MAAGAARD, 2016; SHI et al., 2016]



ARTIFICIAL INTELLIGENCE

The use of Artificial Intelligence is already a reality worldwide, to assist design, construction, and modeling at all stages of a building.

[FARZANEH et al., 2021]

RESEARCH QUESTION & METHODOLOGY

Which architectural design parameters most significantly impact heating and cooling energy demand in residential buildings under Cfa climate conditions?

01

Massive Simulation Campaign

40,000 simulations conducted using EnergyPlus and NSGA-II optimization, yielding 320 Pareto Front envelope solutions.

02

Machine Learning Analysis

Random Forest models developed to predict annual energy consumption with SHAP interpretation for feature importance assessment.

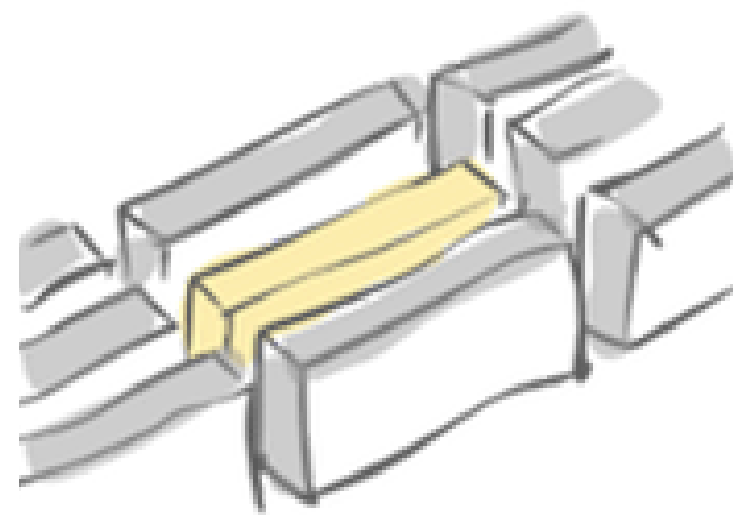
03

Comparative Assessment

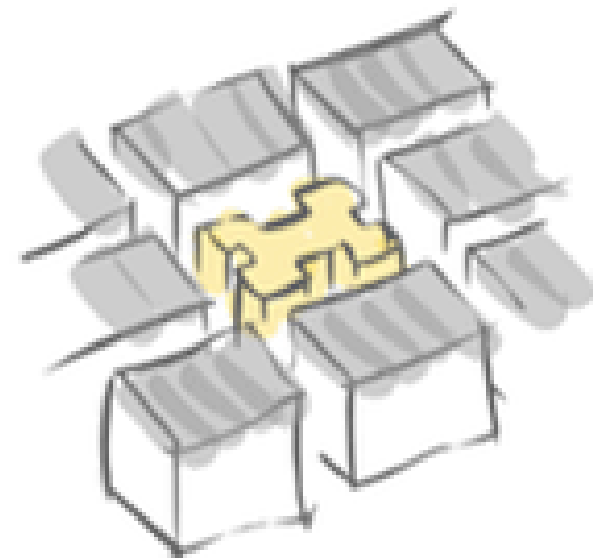
Separate analyses for H and Linear buildings enabling shape-specific sensitivity evaluation.

PASSO FUNDO | BRASIL

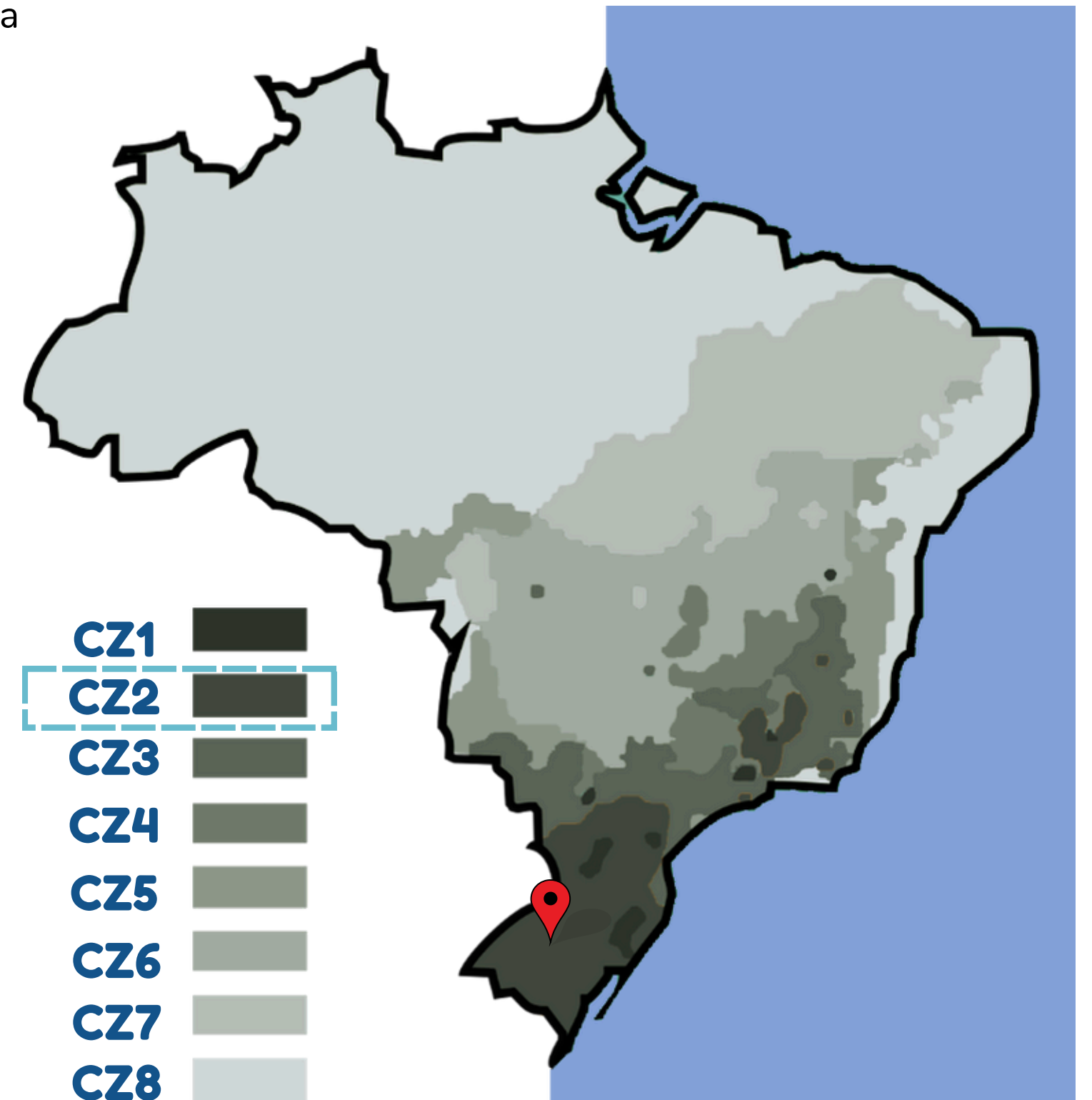
Köppen-Geiger classification: Cfa



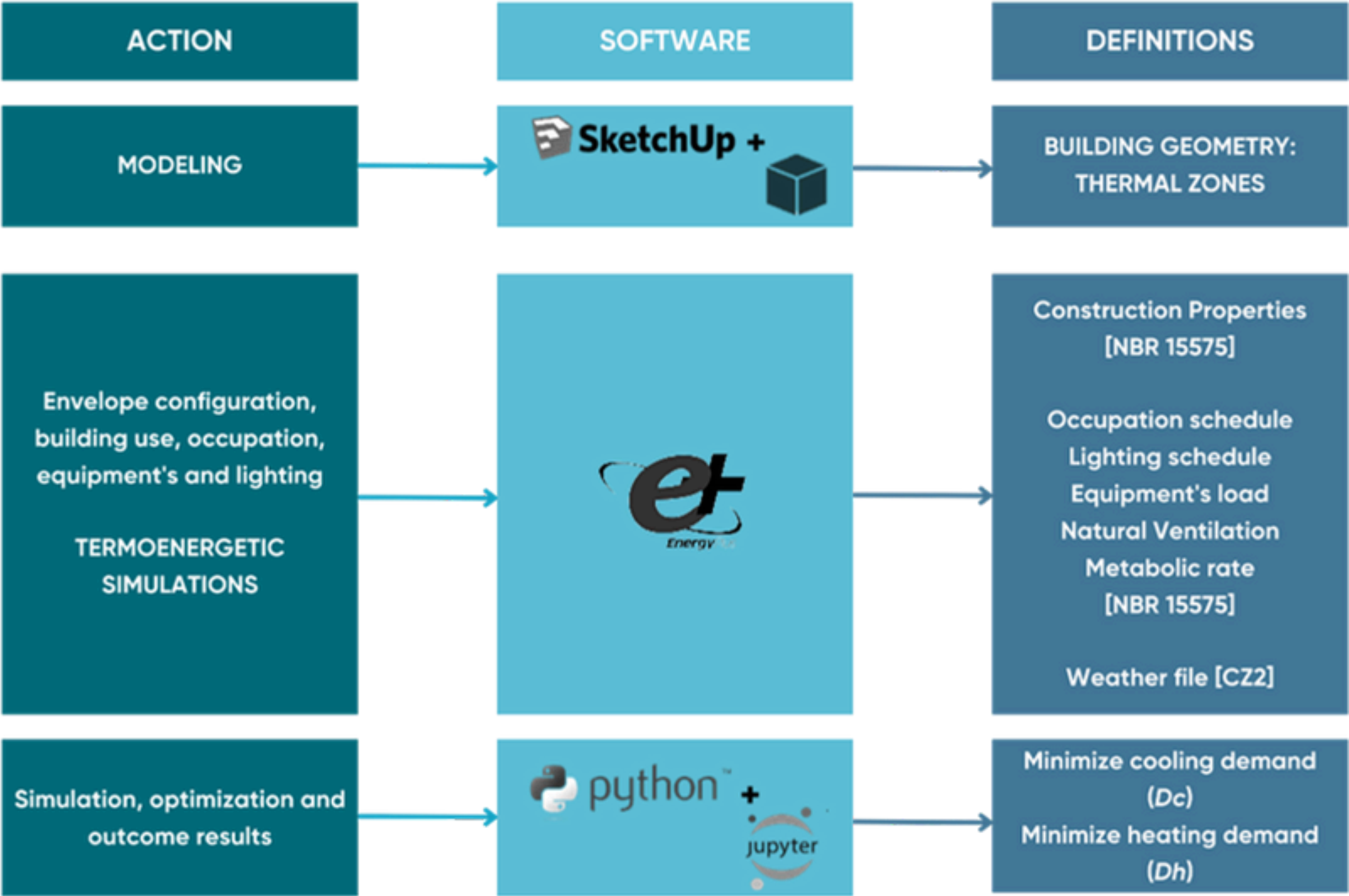
LINEAR BUILDING



H BUILDING



THREE PHASE OPTIMIZATION FRAMEWORK PROCESS

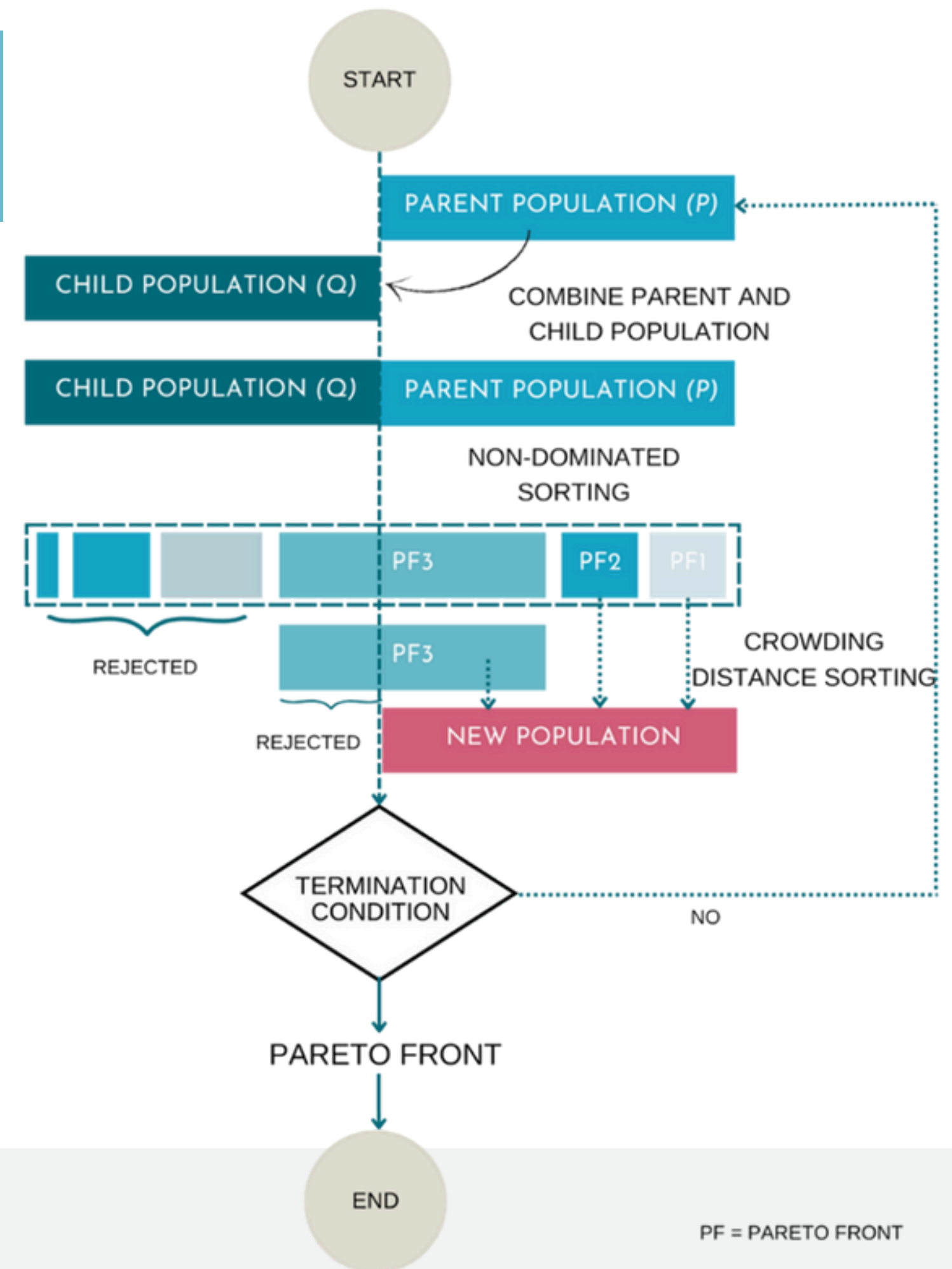


NSGA-II

Non-Dominated Sorting Genetic Algorithm II

It is often considered the best multi-objective genetic algorithm for energy efficiency optimization in buildings [DEB et al., 2002; KHIERI, 2018; ATTIA et al., 2013].

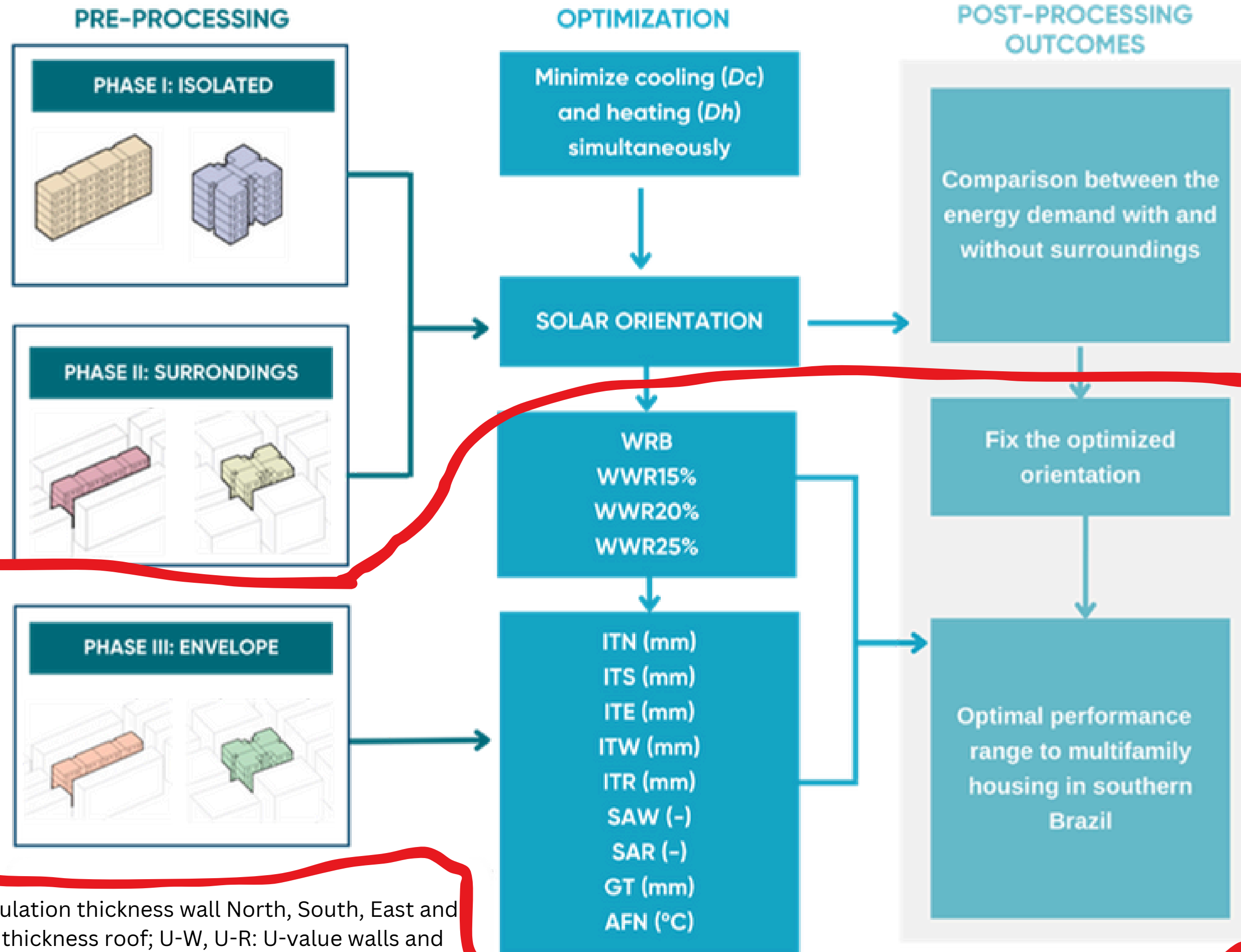
Its efficiency and effectiveness are guaranteed by the elitism mechanism, which preserves the best solutions in each generation, and by the non-dominated sorting technique, which maintains a diverse set of non-dominated solutions in the Pareto space [ALAJMI; WRIGHT, 2014; CHEN; TSAY; NI, 2022; DEB, 2011; EVINS, 2013].



THREE-PHASE OPTIMIZATION FRAMEWORK

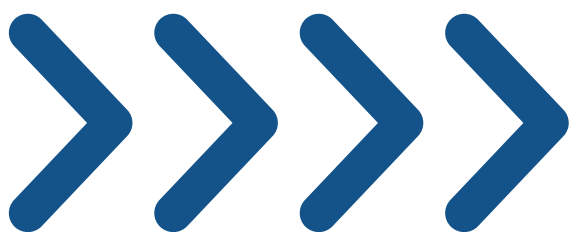
Energy and Buildings
2023 Publication

Energy and Buildings
2024 Submitted

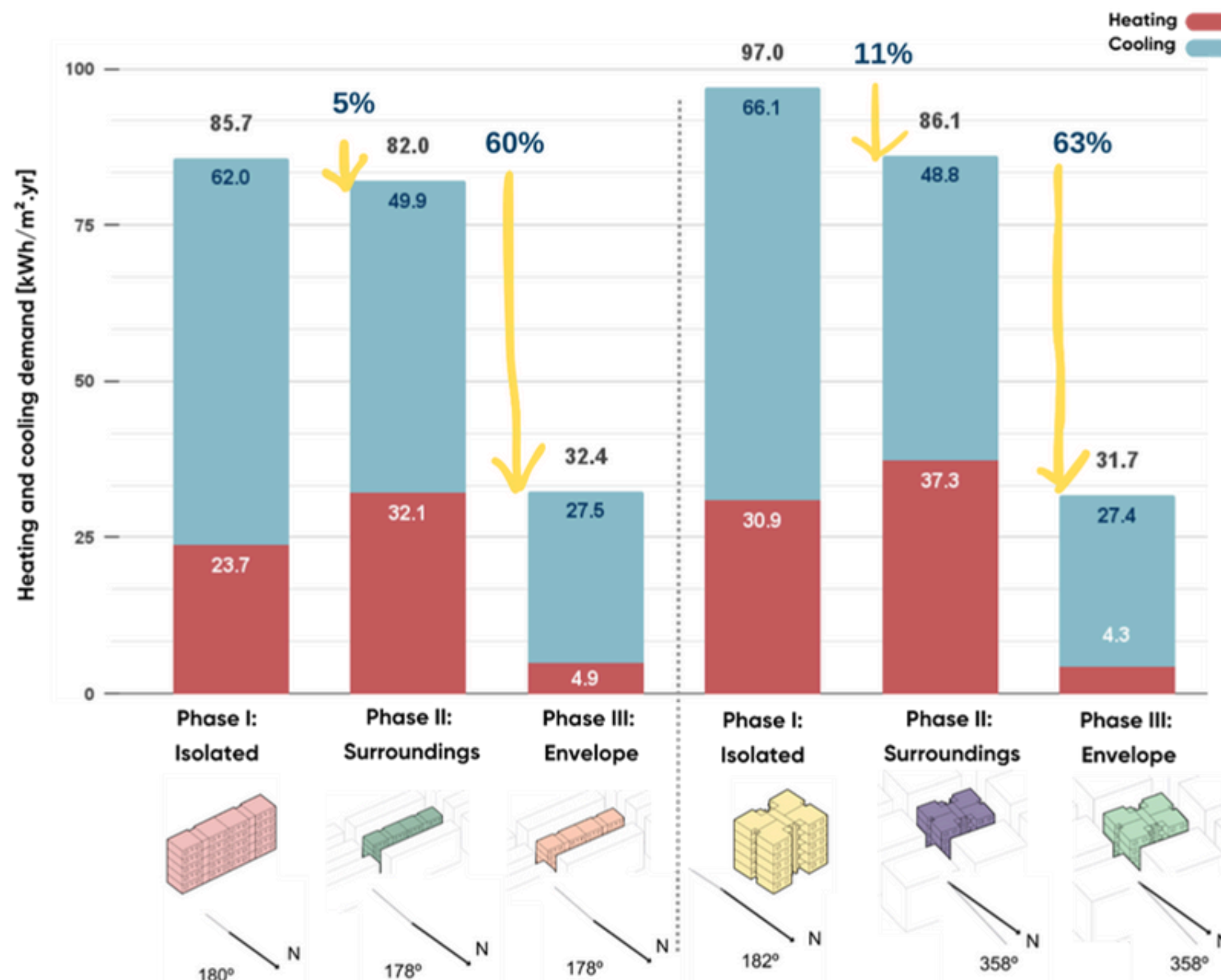


Legend:

ITN, ITS, ITE, ITW: Insulation thickness wall North, South, East and West; ITR: Insulation thickness roof; U-W, U-R: U-value walls and roof; SAW, SAR: Solar absorptance wall and roof; AFN: Natural ventilation Setpoint.



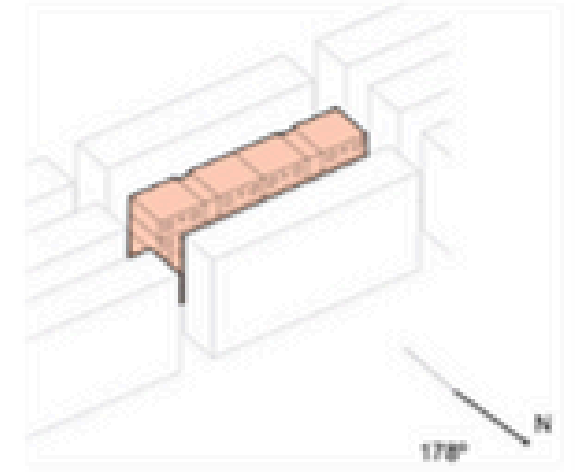
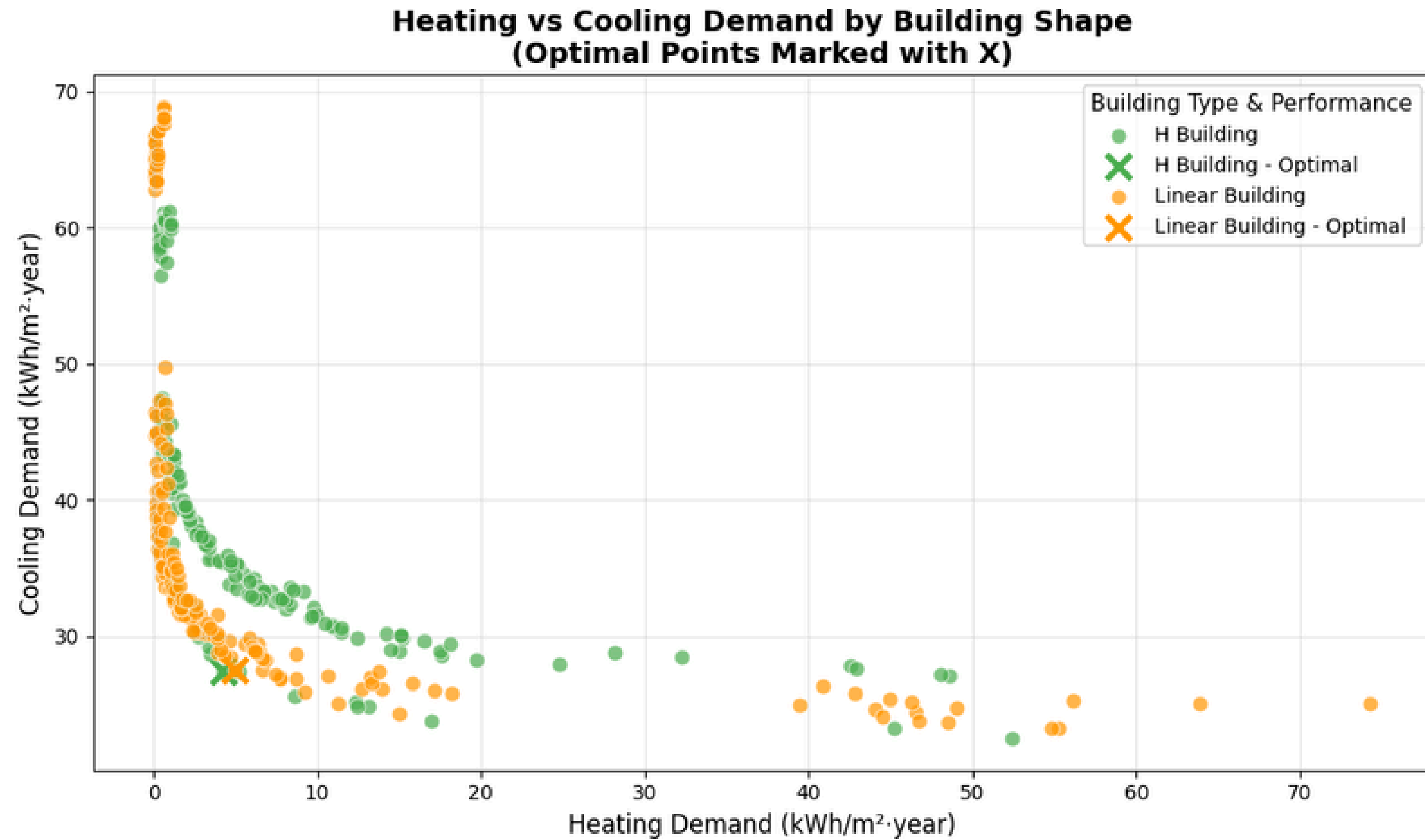
COMPARISON BETWEEN PHASES



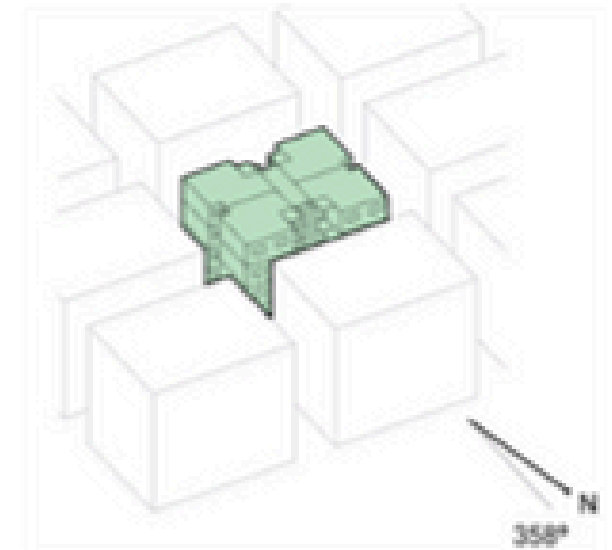
PHASE III

ENVELOPE

INTEGRATED TO SURROUNDINGS



Linear Building

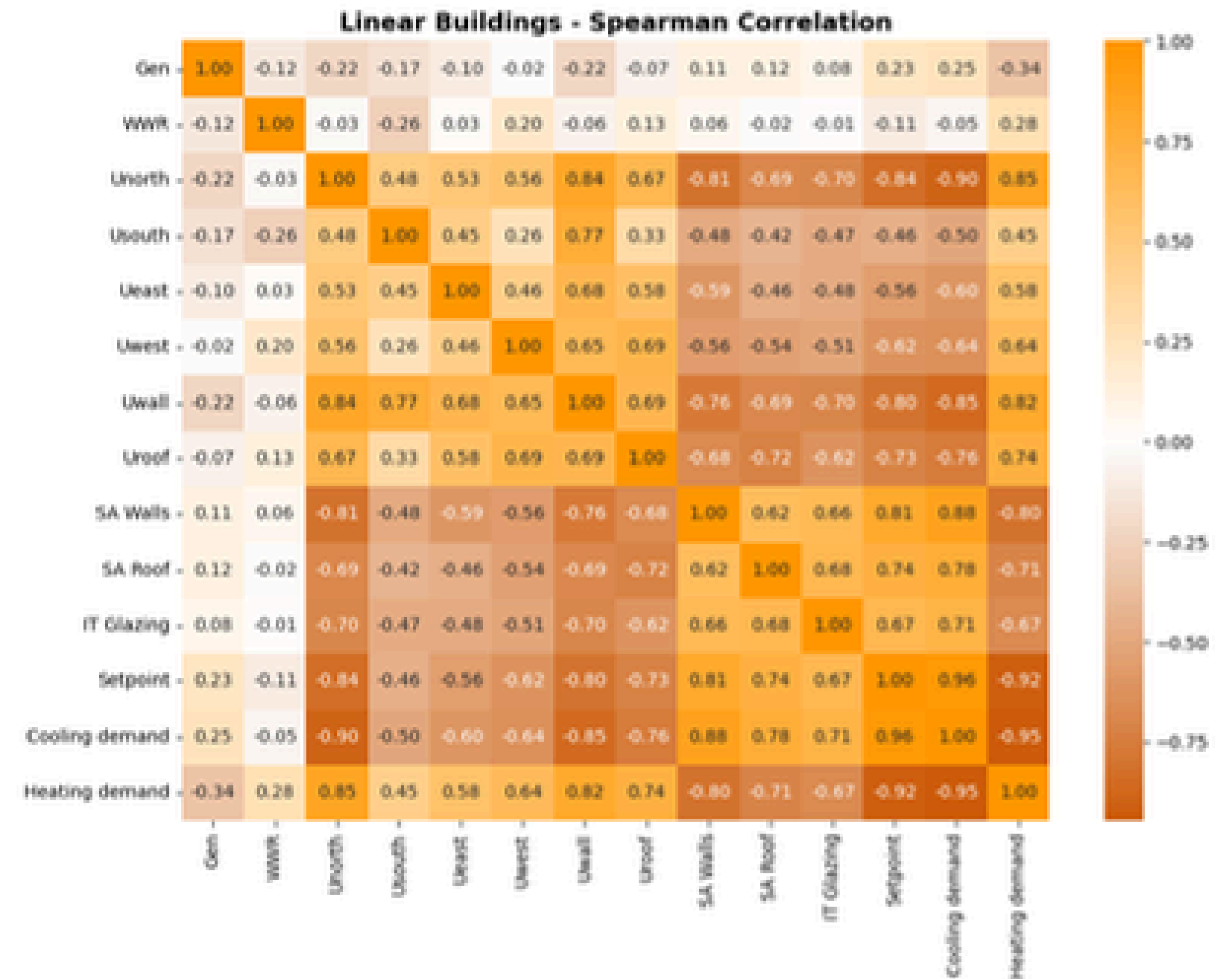
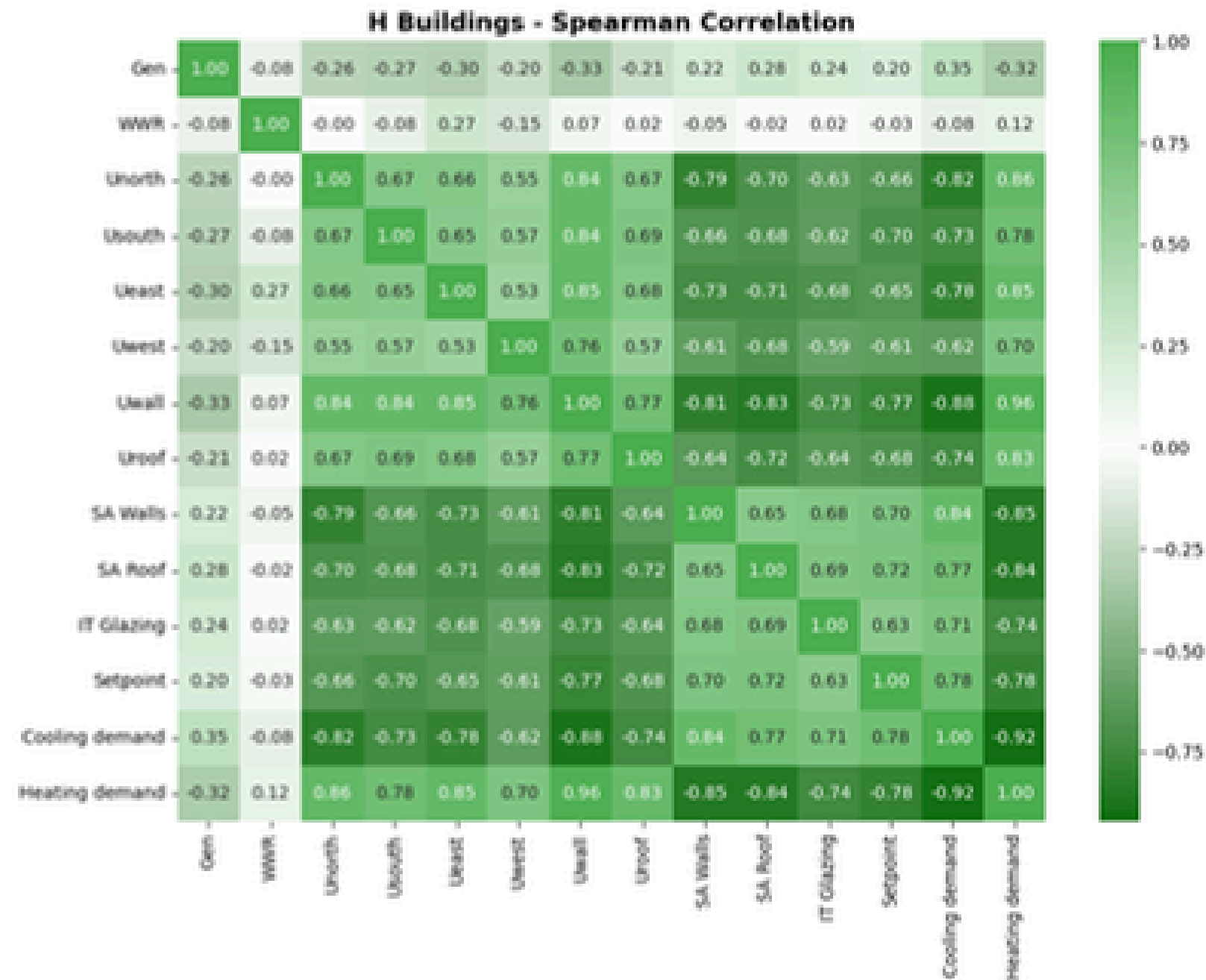


H Building

PHASE III

ENVELOPE

INTEGRATED TO SURROUNDINGS



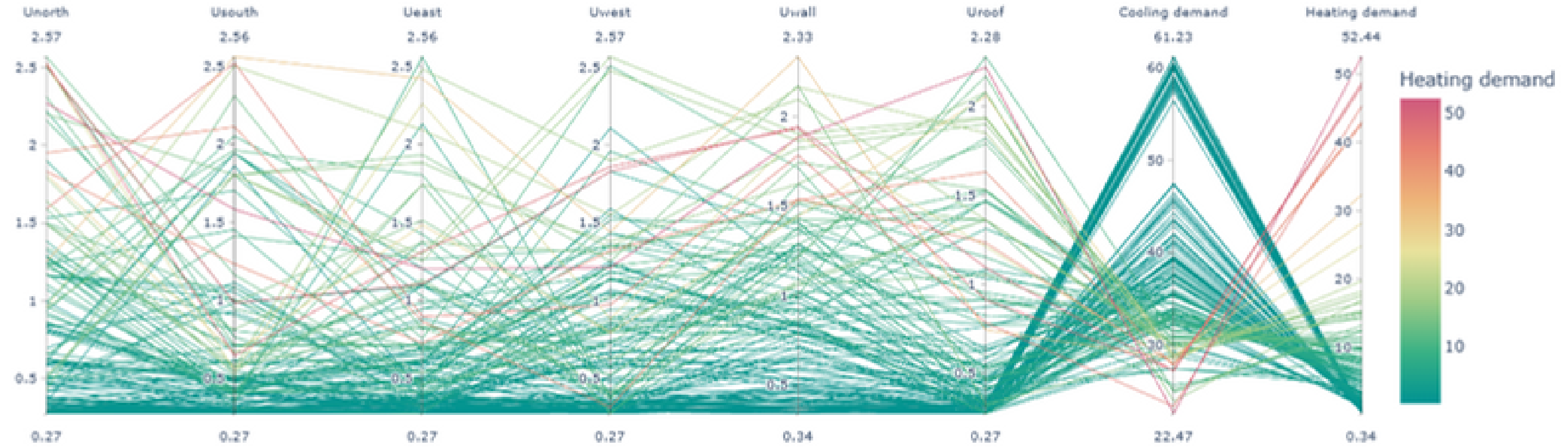
H buildings show that wall and roof insulation strongly controls energy demand, with north wall U-values critically affecting cooling loads, while Linear buildings are even more sensitive to envelope performance, requiring stricter thermal control and reflective surfaces due to their elongated form's increased solar exposure. Both building types benefit from good insulation, but Linear buildings demand more precise passive design strategies, particularly for north façade and roof/wall U-values, to optimize energy efficiency.

PHASE III

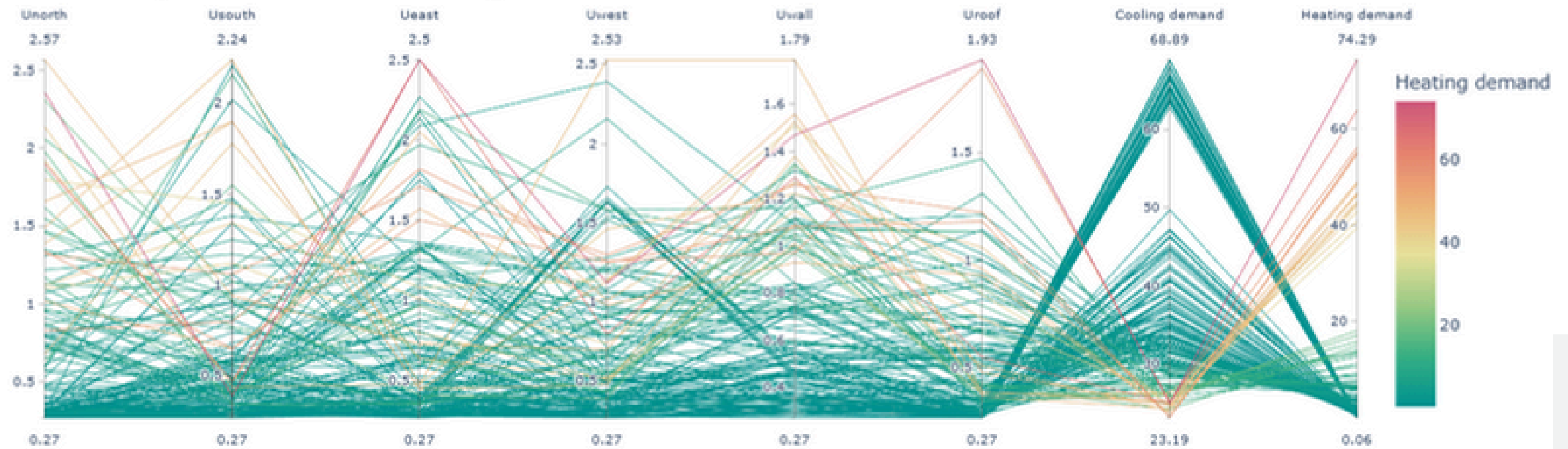
ENVELOPE

INTEGRATED TO SURROUNDINGS

ParallelGroup Coordinates: H Building



ParallelGroup Coordinates: Linear Building

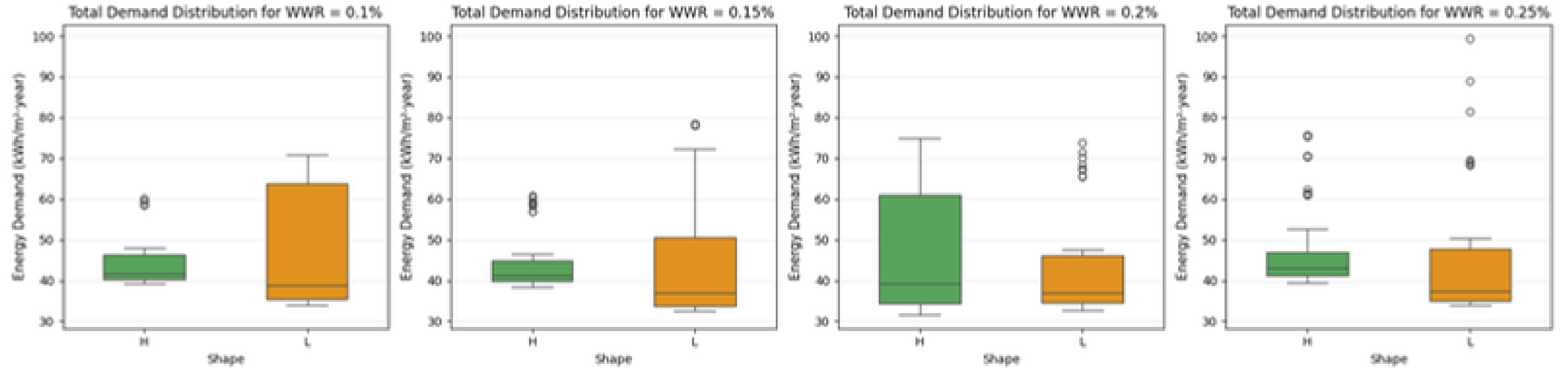


PHASE III

ENVELOPE

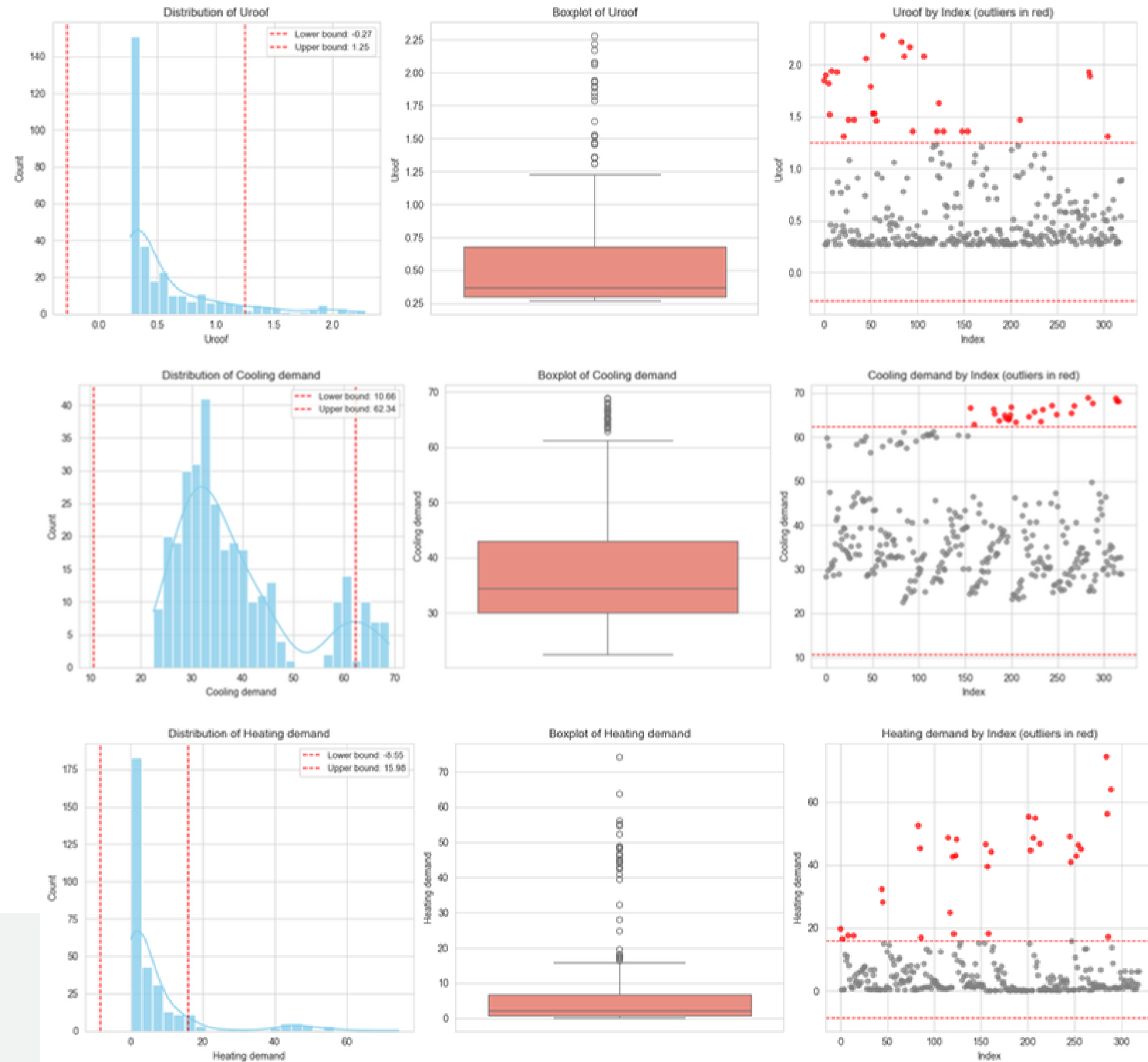
INTEGRATED TO SURROUNDINGS

Comparison of Total Demand by WWR and Shape

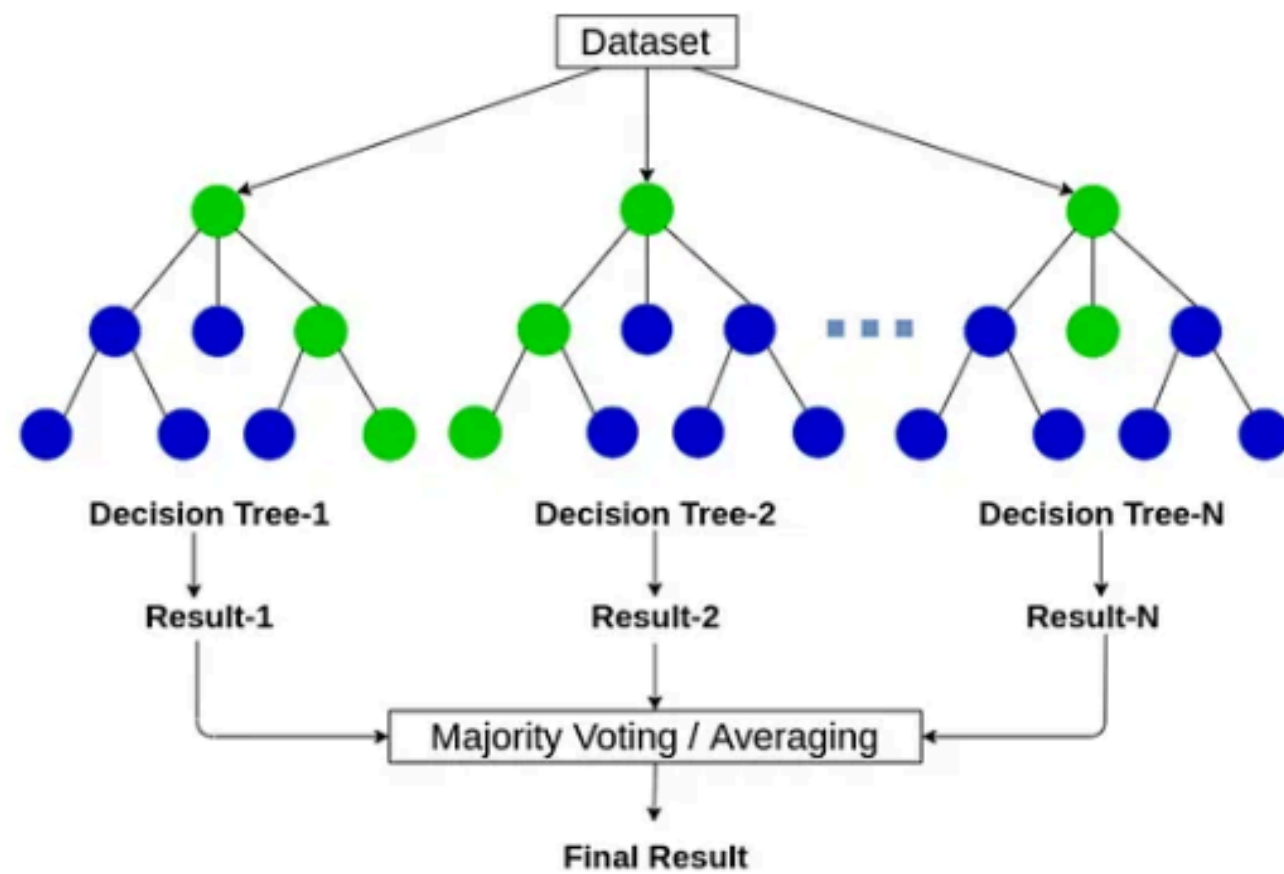


H buildings consistently consume less energy than Linear buildings across all WWR values, with the most significant difference at 20% WWR (38 vs 45 kWh/m²·year), while Linear buildings show higher variability and energy demand, particularly at low WWRs, indicating that H-shaped geometry provides superior thermal performance and energy efficiency.

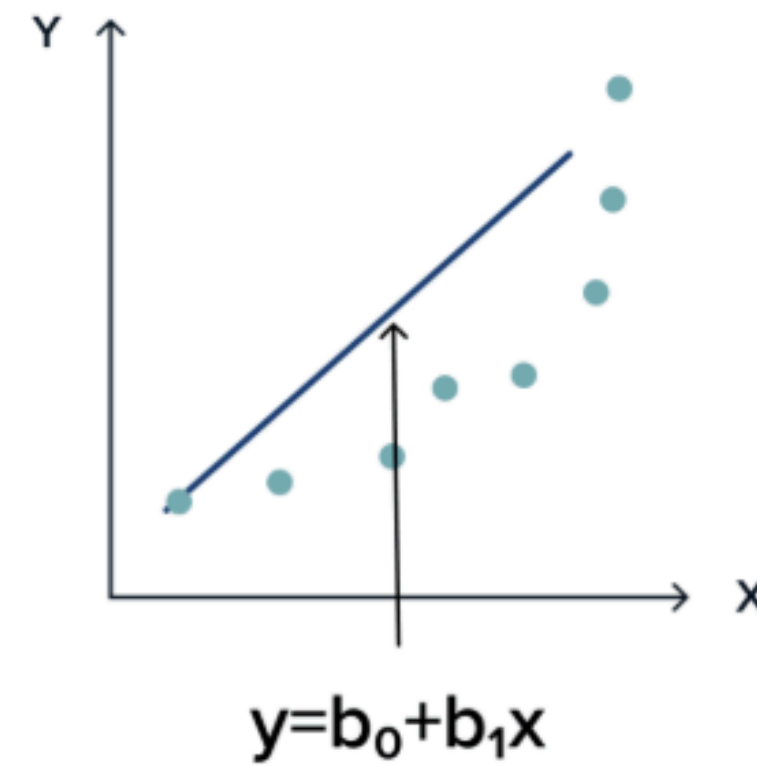
EDA - OUTLIERS



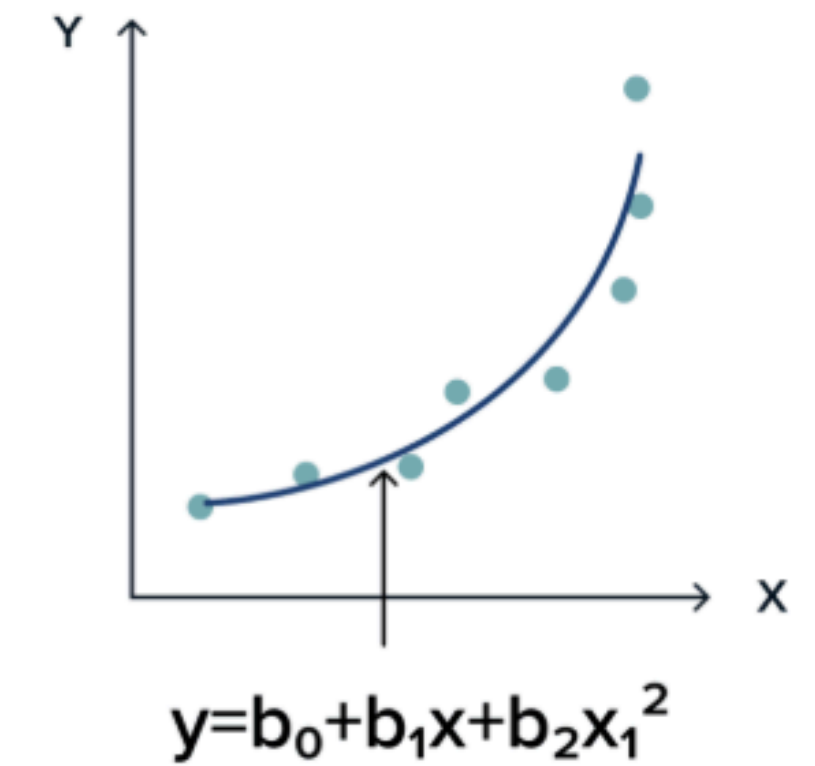
Random Forest



Simple linear model



Polynomial model



Target	Baseline R^2	Poly R^2	ΔR^2	Baseline RMSE	Poly RMSE	Δ RMSE
Cooling	0.898652	0.884159	-0.014493	3.472918	3.751140	0.278221
Heating	0.834020	0.854688	0.020668	3.903315	3.766077	-0.137239

SHAP (SHapley Additive exPlanations) is a unified framework for interpreting machine learning model predictions.

What SHAP Does:

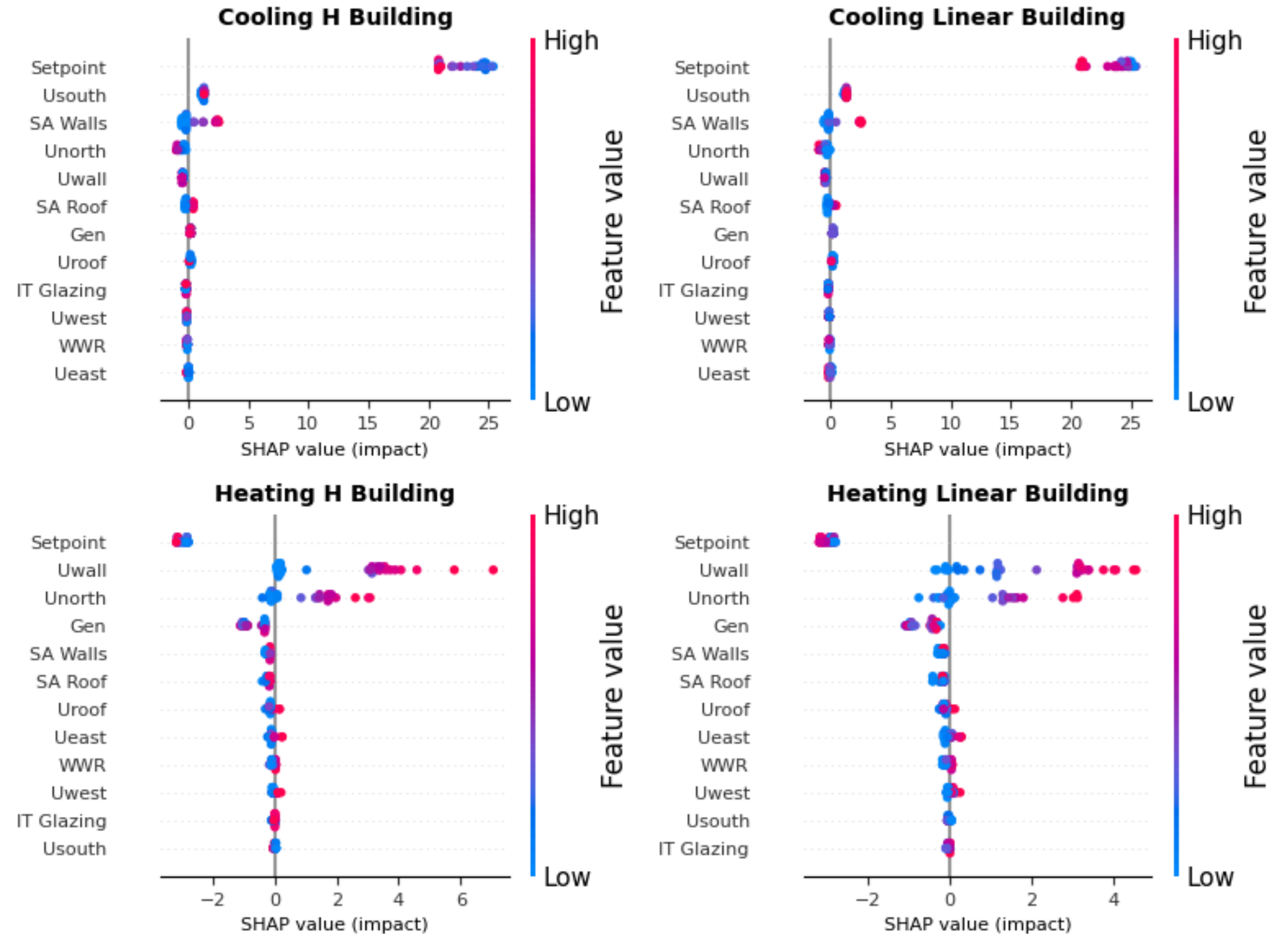
Explains individual predictions

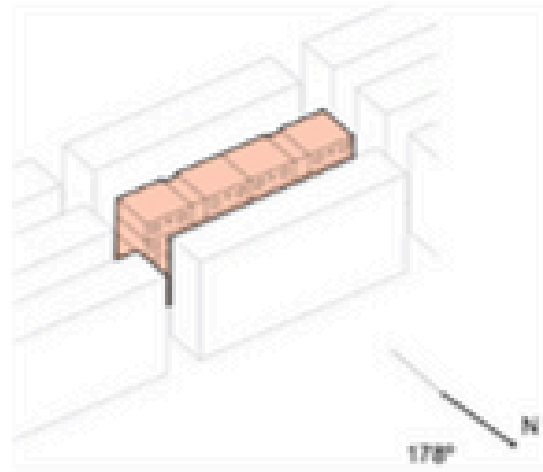
Provides global insights

Works with any model

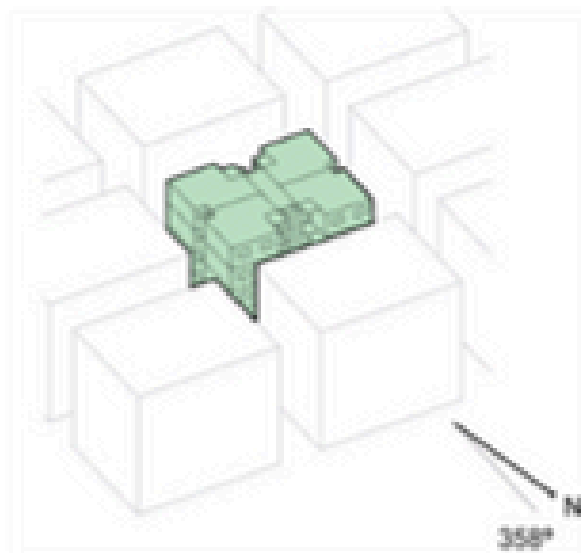
SHAP is based on Shapley values from cooperative game theory: Each feature is treated as a "player" in a game. The "payout" is the model's prediction
Shapley values fairly distribute the contribution among all features

SHAP Feature Importance H vs Linear Buildings





Linear Building



H Building

Shape-Specific Energy Drivers

H Buildings

Cooling: Dominated by operational setpoints and south-facing thermal transmittance (U_{south}), indicating solar exposure and indoor temperature settings are primary contributors.

Heating: Overwhelmingly governed by wall insulation performance (U_{wall}), with roof thermal transmittance (U_{roof}) playing a critical secondary role.

Linear Buildings

Cooling: Window-to-wall ratio (WWR) emerges as most critical factor after setpoint, with stronger dependence on operational parameters.

Heating: Setpoint remains dominant, while physical envelope characteristics like wall insulation play secondary roles compared to H buildings.



CONCLUSIONS

This study demonstrates that while envelope transmittance optimization remains vital for building energy performance, its effectiveness depends highly on building form and occupant-controlled variables like setpoint temperature.

1

Current Achievement

SHAP-based sensitivity analyses provide actionable, interpretable insights supporting morphology and behavior-driven design strategies.

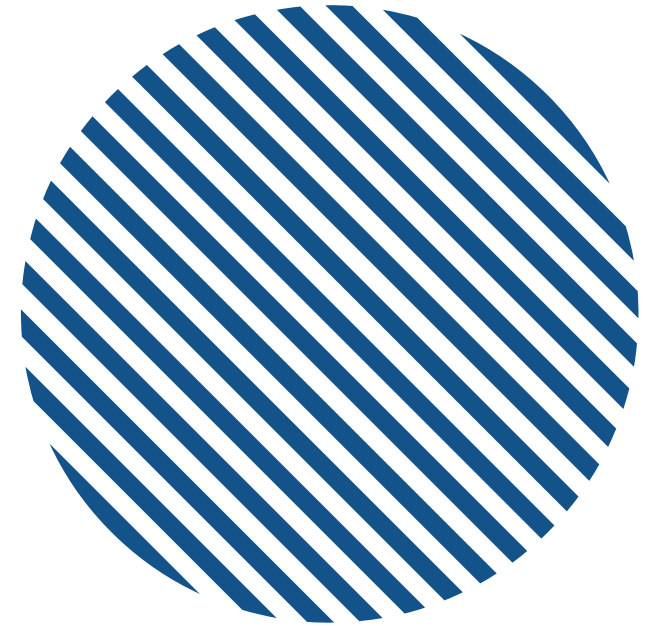
2

Future Research

Extend this approach to varied climates, typologies, and occupancy schedules to further validate and generalize results.

The path to energy-efficient buildings lies not just in advanced materials and systems, but in understanding the complex interplay between architectural form, envelope performance, and human behavior.

THANK YOU / GRÀCIES / GRACIAS / OBRIGADO



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